

4) Intelligent Agents in a smart Home Automation System

An intelligent agent is a system that perceives its environment through sensors and takes actions through actuators to achieve a desired objective.

Sensors may detect

- ⇒ Motions or human presence
- ⇒ Temperature
- ⇒ Light level
- ⇒ Time of day
- ⇒ user activities and preferences

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The agent then controls devices such as lights, fan, air conditioners, heaters, alarms and locks.

There are four important types of intelligent agents:

Simple Reflex Agent

A simple reflex agent chooses an action based only on the current percept using condition-action rules.

It condition → perform action

- ⇒ If motion is detected → turn on the light

If temperature $\rightarrow 28^{\circ}\text{C} \rightarrow$ turn on air conditioner.

→ Model-Based Agent

A model-based agent maintains an internal model/state of the environment. It uses current sensor information together with information about previous situations

Smart home example

Suppose the motion sensor detects that someone entered the house. The agent remembers that the person is probably in the living room and can use this information.

→ Goal Based Agent

A goal-based agent selects actions based on a desired goal

For a smart home, goals could include

⇒ Maintain comfortable temperature

⇒ Reduce electricity consumption

⇒ Keep the house secure

⇒ Turn off unnecessary appliances

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How it works

- ⇒ Start from the initial position
- ⇒ Add possible movements to the priority queue
- ⇒ Select the path with the lowest total cost
- ⇒ Expand the state
- ⇒ continue until the goal is reached

Example

It moving through some maze paths has different costs, UCS ensure that the robot finds the least-cost.

Iterative Deepening search

IDS search repeatedly performs depth-limited search, increasing the depth limit

Depth 0 → Depth 1 → Depth 2 → Depth 3 ...

Advantages

- ⇒ uses very little memory
- ⇒ complete when the branching factor is finite

Disadvantages

- ⇒ Repeats the search at each depths
- ⇒ can perform unnecessary work